

Effects of Interaction Modality and Informational Correctness on Trust Perception in a
Multimodal Conversational System

by
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This thesis is dedicated to my family for their unwavering support, and to my mentors for their guidance and belief throughout this journey.

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Chapter 1 Abstract

Trust is a critical factor in human-robot interaction, particularly in socially interactive systems. Although emotion recognition technologies have advanced substantially, it remains unclear whether variations in expressive modality and informational reliability influence perceived trust during brief interactions. This thesis examines the design and evaluation of a multimodal conversational system that integrates facial and speech-based emotion recognition with adaptive interaction mechanisms. Sixty participants were randomly assigned to one of four conditions in a 2×2 between-subjects design manipulating input modality (voice only vs. voice with facial expressions) and factual correctness (correct vs. intentionally incorrect response). Participants engaged in a structured in-person conversational session involving dialogue, a trivia task with manipulated informational accuracy, and an emotion expression task. Following the interaction, participants completed a questionnaire assessing perceived trust, adaptability, predictability, and emotion recognition accuracy. Two-way analyses of variance revealed no statistically significant main effects or interaction effects of input modality or factual correctness on perceived trust, adaptability, predictability, or accuracy. Correlation analyses indicated positive but non-substantial associations between perceived trust and both perceived adaptability and perceived accuracy. Predictability did not demonstrate a consistent relationship with trust, likely due to limitations associated with measuring the construct using two items. These findings suggest that modest variations in modality and correctness may not produce detectable trust differences in short-term interactions, highlighting the importance of measurement precision in trust research.

1.1 Background and Motivation

Socially interactive systems are increasingly used in education, healthcare, and service settings. In these environments, effectiveness depends not only on task performance but also on users' perceptions of trust and responsiveness. Research in human-robot interaction shows that trust strongly influences user acceptance and willingness to rely on automated systems (Hancock et al., 2011; Lee and See, 2004; Hoff and Bashir, 2015).

Trust in automation refers to a user's willingness to rely on a system based on beliefs about its competence, reliability, and predictability. In interactive systems, trust is shaped not only by accuracy but also by perceived behavioral consistency and social appropriateness (Lee and See, 2004).

Although affective computing enables systems to detect emotions using facial expressions and speech signals, many conversational platforms rely on largely scripted responses. Limited responsiveness to users' emotional states may reduce perceived authenticity and weaken user confidence during interaction. Understanding how perceived adaptability, predictability, and emotion recognition accuracy relate to trust is therefore important for the design of multimodal conversational systems.

1.2 Problem Statement

Although emotion recognition techniques using facial expressions and speech have matured significantly, there remains a lack of robotic systems that effectively integrate emotional feedback into adaptive behavior over time. In particular, companion robots often fail to demonstrate learning from prior interactions, which can negatively affect users' perceptions of trust and predictability (Lee and See, 2004; Hancock et al., 2011).

1.3 Research Objectives

The objectives of this thesis are to:

- Design a human companion robotic system capable of recognizing emotional cues from facial expressions and speech.
- Integrate direct and indirect emotional feedback from human users into the robot's adaptive interaction and behavior-selection framework.
- Evaluate the impact of emotion-informed adaptation on perceived trust, accuracy, adaptability, and predictability.

1.4 Scope and Contributions

This thesis focuses on short-term, in-person human-robot interaction using a multimodal conversational system that integrates speech interaction with facial and vocal emotion recognition capabilities. In this study, adaptive behavioral mechanisms refer to rule-based adjustments of conversational responses during the interaction session based on detected emotional cues and contextual information. These mechanisms do not involve long-term learning, retraining of machine learning models, or persistent memory across sessions. The study evaluates user perceptions of emotion-informed response behavior within a structured experimental interaction rather than algorithmic optimization or longitudinal trust development. The primary contributions of this thesis are as follows. Firstly, the design and implementation of a multimodal conversational system integrating speech interaction with facial and vocal emotion sensing. Secondly, a controlled 2×2 experimental evaluation examining the effects of input modality and informational correctness on user perceptions. Thirdly, an empirical examination of associations between perceived adaptability, predictability, emotion recognition accuracy, and trust in a structured interaction context.

1.5 Thesis Organization

This thesis is organized into five chapters. Chapter 1 introduces the research problem, motivation, objectives, and hypotheses, and outlines the scope and contributions of the study. Chapter 2 presents a review of relevant literature on trust in human-robot interaction, including conceptual models of trust, measurement approaches, and factors such as adaptability, predictability, and emotion recognition. It concludes by identifying the research gap addressed in this work. Chapter 3 describes the methodology, including the design and implementation of the multimodal conversational system, experimental design, participant details, data collection procedure, and analysis methods. Chapter 4 presents the results of the study, including descriptive statistics, reliability analysis, analysis of variance (ANOVA), and correlation analyses related to the research hypotheses. Chapter 5 discusses the findings in the context of existing literature, interprets the results, outlines limitations, and suggests directions for future work, followed by the overall conclusion of the study.

Chapter 2 Literature Review

2.1 Trust in Human-Robot Interaction

Trust is a central construct in human-robot interaction and plays a critical role in determining whether users are willing to rely on, cooperate with, and accept interactive systems. Prior research characterizes trust as a multidimensional and context-dependent construct reflecting a user's willingness to accept vulnerability based on expectations of a system's competence, reliability, and predictability (Hancock et al., 2011; Hoff and Bashir, 2015; Lee and See, 2004; Mayer et al., 1995). Trust is commonly defined as a user's willingness to rely on a system despite uncertainty or risk. It is shaped by perceptions of the system's competence, reliability, and predictability, and varies depending on the context and interaction. In human-robot interaction, trust is particularly important because interactive systems increasingly operate in close physical, social, and decision-making proximity to users. Systematic reviews indicate that trust is influenced by system-related factors such as reliability and behavioral consistency, as well as human-related factors such as prior experience and expectations (Hoff and Bashir, 2015). For instance, in driving automation research, participants trusted adaptive cruise control systems more when the system maintained steady, predictable behavior across repeated interactions. Conversely, users with prior negative experiences or low expectations of automation exhibited reduced trust – even when the system performed correctly.

Research also suggests that trust can evolve during interaction as users update their beliefs about system performance (Kohn et al., 2021). Specifically, trust is shaped by moment-to-moment factors such as the system's response accuracy, consistency across repeated actions, transparency in decision-making, and its ability to appropriately adapt to user input or context. Positive interactions – such as correct, timely, and contextually appropriate responses – tend to

reinforce trust, whereas errors, unpredictable behavior, or mismatches between user expectations and system actions can rapidly degrade it. Although trust development may occur dynamically, many experimental studies assess trust using post-interaction self-report measures to capture users' overall evaluation following an interaction session. The present study adopts this approach and examines perceived trust after a structured interaction with the multimodal conversational system.

2.2 Measurement of Trust in Robotic and Artificial Systems

Accurate measurement of trust is essential for empirical research in human-robot interaction. Trust is commonly assessed using self-report questionnaires administered immediately after an interaction session, particularly in short-duration laboratory studies where behavioral trust measures may be difficult to interpret (Lee and See, 2004; Hoff and Bashir, 2015). These questionnaires typically capture users' overall judgments about a system's reliability, competence, and willingness to rely on it in future interactions. In this study, these dimensions are directly operationalized through measures of perceived trust, adaptability, predictability, and accuracy, which reflect key components identified in prior trust literature. The experimental manipulations of input modality and informational correctness are designed to influence these perceptions by altering how the system communicates and how reliably it performs. As a result, the selected measures allow for a structured evaluation of how these design factors shape users' trust-related judgments in line with established theoretical frameworks.

Several validated instruments exist for measuring trust in automation and related technologies. However, many established scales were developed for task-oriented automation contexts and include items that assume specific system functions, long-term exposure, or high-

stakes operational settings. In socially interactive systems, trust judgments may also reflect perceived responsiveness and interaction quality, which are not always fully captured by general automation-focused instruments (Hoff and Bashir, 2015; de Visser et al., 2020). For this reason, many HRI studies adapt existing trust items or use short, context-specific questionnaires that match the interaction scenario and the constructs under investigation (Hoff and Bashir, 2015; Hancock et al., 2011). Consistent with this practice, the present study used a brief post-interaction questionnaire designed to measure perceived trust alongside related constructs of adaptability, predictability, and perceived emotion recognition accuracy. These constructs were included not only to examine their relationship with trust, but also to assess whether the experimental manipulations influenced participants' perceptions in the intended manner. Specifically, informational correctness was expected to affect perceived accuracy and predictability, while input modality was expected to influence perceived adaptability and aspects of predictability through the presence of additional communicative cues. This approach allows trust to be assessed in a manner that is aligned with the system's interaction context while also providing a basis for interpreting how the manipulated system features shape users' perceptions.

2.3 Adaptability and Trust

Adaptability refers to a system's capacity to modify its behavior in response to user cues or interaction context. Prior work in human-robot interaction shows that responsive and context-sensitive behavior influences perceptions of competence and trust (Sebo et al., 2018; de Visser et al., 2020). In trust in automation theory, users update their trust judgments based on observed system behavior over time, particularly when the system demonstrates responsiveness to changing conditions or appropriately manages errors (Lee and See, 2004). Systems that adapt

following user feedback or performance failure are more likely to be perceived as competent and capable of maintaining reliable performance.

Empirical evidence supports this relationship. Hancock et al. (2011), in a meta-analysis of trust in human-robot interaction, identified robot performance characteristics as among the strongest predictors of trust, including reliability, error management, and behavioral responsiveness. de Visser et al. (2016) experimentally demonstrated that systems that provided adaptive responses following errors were more effective at repairing degraded trust compared to systems that maintained fixed behavior. In social interaction contexts, Sebo et al. (2018) showed that conversational systems that adjusted their verbal and nonverbal behaviors based on user engagement cues were perceived as more socially intelligent and trustworthy than conversational systems that followed predetermined scripts. These findings suggest that perceived adaptability strengthens trust because it signals competence, responsiveness, and alignment with user expectations.

2.4 Predictability and Behavioral Consistency

Predictability refers to the extent to which a system's behavior is consistent, understandable, and aligned with user expectations (Lee and See, 2004; Hoff and Bashir, 2015). Trust models in automation consistently identify predictability and reliability as core determinants of calibrated trust. When users can anticipate how a system will behave, they are more likely to form accurate mental models of the system's functioning. A mental model refers to a user's internal understanding of how a system operates. Accurate mental models reduce uncertainty and increase willingness to rely on the system.

Empirical research reinforces this connection. Hancock et al. (2011) reported that perceived reliability and consistency were among the strongest predictors of trust across human-

robot interaction studies. Hoff and Bashir (2015) describe predictability as a key factor influencing dynamic trust development, particularly during repeated interaction. de Visser et al. (2020) further show that consistent system performance supports stable trust calibration, whereas unpredictable behavior leads to trust degradation even when objective performance remains adequate. In socially interactive systems, inconsistent or incoherent responses may undermine user confidence because they disrupt expectations about how the system will behave. Therefore, higher perceived predictability is expected to be associated with higher trust.

2.5 Emotion Recognition and Social Trust

Emotion recognition enables interactive systems to interpret and respond to users' affective states, which can support more natural and socially appropriate interaction. In human-robot interaction, affective responsiveness has been linked to perceptions of social intelligence and competence (Dautenhahn, 2019). When a system appears to correctly identify and respond to emotional cues, users are more likely to perceive it as understanding and responsive, which contributes to positive evaluations of interaction quality.

Trust theory suggests that perceptions of competence and responsiveness influence overall trust judgments (Lee and See, 2004; Hoff and Bashir, 2015). In socially interactive contexts, emotion recognition may serve as a signal of competence because it indicates that the system can interpret relevant social information and adjust behavior accordingly. Empirical research supports this relationship. Langer et al. (2019) found that socially expressive conversational system behaviors that were aligned with users' emotional states increased perceptions of social presence and trustworthiness compared to neutral or mismatched responses. Similarly, studies of emotionally responsive agents show that systems perceived as sensitive to user affect are rated as more trustworthy and socially intelligent than systems that ignore

emotional cues (de Visser et al., 2020). These effects are particularly pronounced in interaction contexts where users expect relational engagement rather than purely task-based performance.

Meta-analytic findings further indicate that conversational system performance factors related to perceived competence and social appropriateness significantly predict trust in human-robot interaction (Hancock et al., 2011). Because accurate emotion recognition enhances perceptions of understanding and appropriate responsiveness, it is expected to contribute positively to trust judgments. Together, this literature suggests that when users perceive a system as accurately identifying and responding to their emotional states, they are more likely to view it as competent and responsive, thereby strengthening trust.

2.6 Summary and Research Gap

The reviewed literature demonstrates that trust in human-robot interaction is a multidimensional construct influenced by conversational system behavior, emotional responsiveness, and human characteristics (Hancock et al., 2011; Hoff & Bashir, 2015). While prior work has examined adaptability, predictability, and trust measurement (Muir, 1994; Lee & See, 2004; de Visser et al., 2018), fewer studies integrate these factors within a unified companion conversational system framework that incorporates multimodal emotion recognition (Breazeal, 2003; Tapus et al., 2007). Furthermore, existing research primarily focuses on industrial or task-oriented contexts, leaving a gap in understanding how adaptive, emotionally responsive robots influence trust in social and companion-oriented interactions. This thesis addresses this gap by empirically examining how adaptability, predictability, and accuracy influence perceived trust in a companion robotic system.

2.7 Hypotheses

H1: Input Modality and Trust

Participants interacting with the multimodal system (voice plus facial expressions) will report higher perceived trust compared to those interacting with the voice-only system. Input modality was manipulated to vary the level of social and perceptual cues available during interaction. The inclusion of facial expressions alongside voice is expected to enhance communicative richness, increase perceived social presence, and make the system appear more responsive and engaging. These additional cues may lead users to infer greater system capability and attentiveness, thereby increasing perceived trust.

H2: Informational Correctness and Trust

Participants exposed to correct system responses will report higher perceived trust compared to those exposed to intentionally incorrect responses. Informational correctness was manipulated to influence perceived system reliability and competence. Correct responses are expected to align with users' expectations and reinforce confidence in the system, whereas incorrect responses may violate expectations and reduce perceived reliability. Because trust in automation is strongly linked to perceptions of system accuracy and dependability, differences in informational correctness are expected to directly influence trust judgments.

H3: Interaction Effect

There will be an interaction between input modality and informational correctness such that the positive effect of multimodal input on perceived trust will be stronger when the system provides correct information than when it provides incorrect information. Specifically, when the system delivers accurate information, the presence of additional social cues may further enhance trust by reinforcing perceptions of competence and engagement. In contrast, when the system provides incorrect information, these cues may be insufficient to compensate for reduced reliability, thereby weakening the effect of modality on trust.

H4: Adaptability and Trust

Higher perceived adaptability of the multimodal conversational system is associated with higher levels of perceived trust. Adaptability refers to the extent to which participants perceived the system as adjusting its responses in a context-sensitive manner during interaction. Adaptability was measured using a two-item post-interaction self-report scale assessing whether the system appeared responsive rather than rigidly scripted.

H5: Predictability and Trust

Higher perceived predictability of the multimodal conversational system's responses is associated with higher levels of perceived trust. Predictability refers to perceived behavioral consistency and coherence, meaning whether responses were understandable and whether participants felt they could anticipate the system's behavior. Predictability was assessed using two post-interaction questionnaire items. Because these items showed low internal consistency, they were analyzed individually rather than combined into a composite score.

H6: Emotion Recognition Accuracy and Trust

Higher perceived emotion recognition accuracy is associated with higher levels of perceived trust. Emotion recognition accuracy refers to participants' perceptions that the system correctly identified and responded to emotional cues in speech and facial expression. This construct was measured using a three-item post-interaction self-report scale evaluating perceived emotional understanding and response appropriateness.

Chapter 3 Methodology

This study employed a quantitative, questionnaire-based research design to examine factors influencing trust in human-robot interaction. The primary objective was to evaluate how perceived adaptability, predictability, and accuracy of a companion conversational system relate to users' perceived trust following an interaction. Specifically, the study used a 2×2 between-subjects design in which input modality and informational correctness were experimentally controlled. Input modality was manipulated by comparing a voice-only interaction condition with a multimodal condition that included both voice and animated facial expressions. Informational correctness was manipulated by varying whether the system provided accurate or intentionally incorrect responses during the interaction. Data were collected using a post-interaction survey administered immediately after participants completed an in-person interaction with the conversational system. The study followed a cross-sectional design, with each participant completing a single interaction session followed by the questionnaire. All data were self-reported, and no behavioral or physiological measures were collected. This design was selected to capture participants' immediate perceptions of trust in the conversational system's behavior.

3.1 Materials

The companion conversational system used in this study consisted of a laptop-based interactive platform augmented with multimodal-sensing hardware. The system integrated visual and auditory input processing with a conversational interface and expressive visual output. The conversational system was implemented using the following hardware components:

- A laptop computer serving as the central processing unit and display interface.

- A Luxonis OAK-D camera used for real-time facial expression detection and visual input processing.
- A ReSpeaker microphone array used to capture high-quality speech input and enable voice emotion recognition.

The OAK-D camera provided RGB video input for facial expression analysis, while the ReSpeaker array captured speech signals for speech recognition and vocal emotion inference. Both modalities were processed in real time during interaction. Participants interacted with the system through spoken conversation. The system responded using synthesized speech output generated through a text-to-speech module. The laptop screen displayed:

1. A real-time transcript of the ongoing conversation between the participant and the conversational system.
2. A visual face representation that animated while the conversational system spoke.

3.2 Configuration

The conversational system was implemented using lightweight, pre-trained models selected to ensure real-time performance under limited computational resources. Speech-to-text (STT) processing was performed using the *Whisper Tiny.en* model. This model was chosen due to its low computational overhead and ability to provide reliable real-time transcription for English speech, making it suitable for interactive settings where latency must be minimized. Vocal emotion recognition (VER) was implemented using *DeepSpectrumLite*, which extracts spectrogram-based features and applies a deep neural network for emotion classification. This model was selected for its optimized and efficient architecture, enabling real-time inference while maintaining reasonable accuracy in emotion detection. Facial emotion recognition (FER) was performed using the *DepthAI* framework on the Luxonis OAK-D camera. This approach

allows on-device inference, significantly reducing processing load on the central system and minimizing latency, which is critical for maintaining smooth multimodal interaction. The conversational response generation module was powered by the *Mistral 7B Instruct* large language model, deployed locally. This model was chosen as a balance between language generation capability and computational feasibility, allowing context-aware responses without relying on cloud-based services, thereby improving responsiveness and system independence. Text-to-speech (TTS) output was generated using a lightweight speech synthesis module *pyttsx3* to ensure low-latency audio feedback. Visual emotional feedback was implemented using a symbolic facial representation, where predefined emoji-based expressions were mapped to detected emotional states. This design choice avoided the computational cost of generative talking-face models and ensured consistent, interpretable visual feedback, while allowing controlled evaluation of multimodal interaction effects. Overall, the system design reflects a deliberate trade-off between model complexity and real-time performance, prioritizing responsiveness, stability, and seamless integration across multimodal components.

Figure 1
Screenshot of the UI

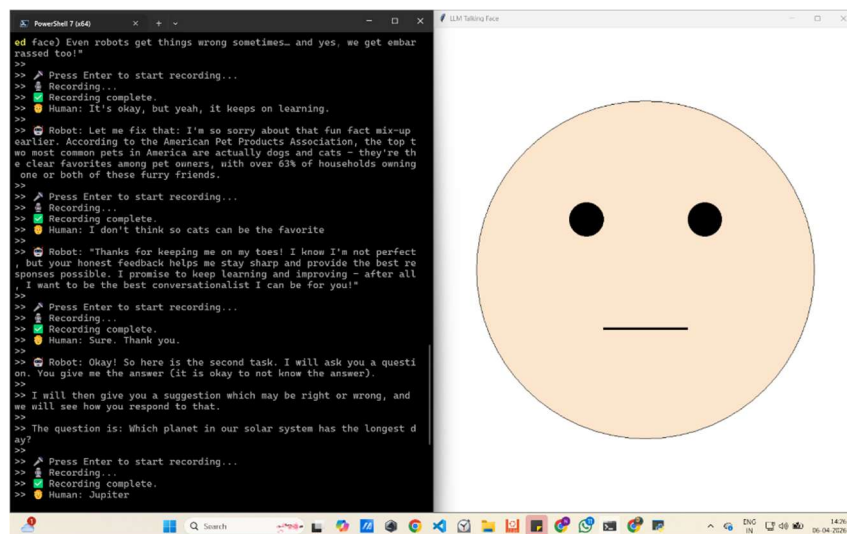
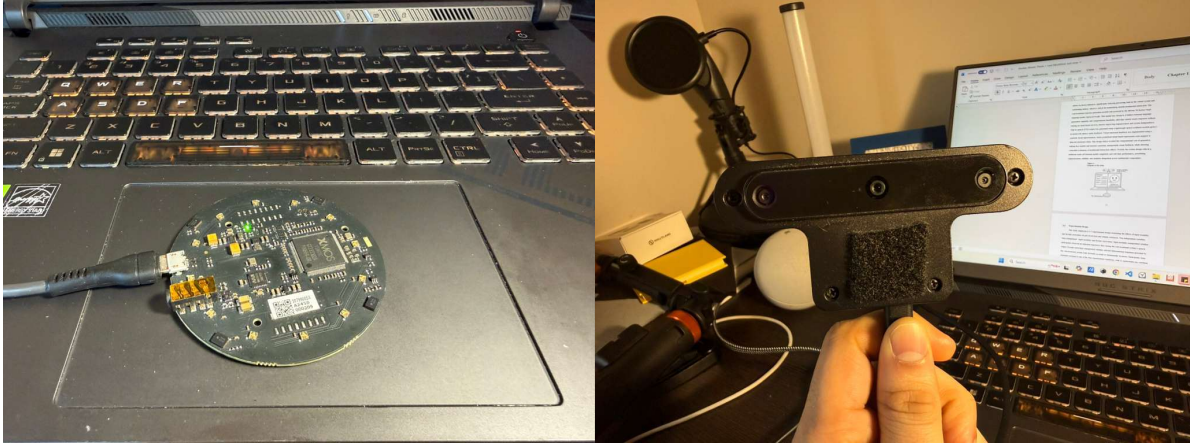


Figure 2
(a) ReSpeaker MicArray v2.0, (b) Luxonis OAK-D depth-camera



3.3 Experimental Design

This study employed a 2×2 experimental design examining the effects of input modality and factual correctness on perceived trust and related constructs. Two independent variables were manipulated - input modality and factual correctness. Input modality manipulated whether participants observed an animated expressive face during the conversational system's speech output. Factual correctness manipulated whether selected informational responses provided by the conversational system were factually accurate or intentionally incorrect. Participants were randomly assigned to one of the four experimental conditions, with 15 participants per condition ($N = 60$ total):

1. Voice only, true factual response ($n = 15$)
2. Voice only, false factual response ($n = 15$)
3. Voice with facial expressions, true factual response ($n = 15$)
4. Voice with facial expressions, false factual response ($n = 15$)

This design allowed examination of the main effect of input modality, and the main effect of factual correctness, the interaction effect between modality and correctness. This experimental

manipulation was designed to examine whether input modality and informational correctness influence participants' perceptions of trust and related constructs. In addition to examining experimental condition effects using analysis of variance (ANOVA), the study also tested associational hypotheses concerning the relationships between perceived adaptability, predictability, emotion recognition accuracy, and trust. These hypotheses were evaluated using correlation analyses across all participants.

3.4 Participants

Participants were 60 individuals assigned to one of four experimental conditions in a 2×2 between-subjects design, resulting in approximately 15 participants per condition. The sample consisted primarily of young adults. Most participants were between 18 and 24 years of age (60.7%, $n = 36$), followed by 25 to 34 years (32.8%, $n = 20$). Smaller proportions were aged 35–44 years (1.6%, $n = 1$) or 45 years and older (4.9%, $n = 3$).

With respect to gender identity, 66.6% of participants identified as male ($n = 40$) and 33.3% identified as female ($n = 20$). The ethnic composition of the sample was predominantly Asian (63.9%, $n = 39$), followed by White (19.7%, $n = 12$), Black or African American (6.6%, $n = 4$), Hispanic or Latino (3.3%, $n = 2$), American Indian or Alaska Native (3.3%, $n = 2$), and Middle Eastern or North African (3.3%, $n = 2$).

Participants were primarily university students (81.66%, $n = 49$). Most students were Master's students (55.7%, $n = 3$), followed by undergraduate students (34.4%, $n = 21$), doctoral students (3.3%, $n = 2$), and other academic or professional categories (6.6%, $n = 4$). Engineering and computing-related disciplines were strongly represented.

Participants reported varying levels of familiarity with artificial intelligence and robotics, with 73.7% ($n = 44$) indicating moderate to high familiarity. Prior interaction experience with

robots varied, with 36.1% (n = 22) reporting no prior interaction, 29.5% (n = 18) reporting some interaction, and 34.4% (n = 21) reporting prior experience interacting with robots. Self-reported technical proficiency was generally high, with 96.7% (n = 58) describing themselves as somewhat tech-savvy or tech-savvy.

3.5 Measures

Trust

Trust refers to the extent to which participants perceived the multimodal conversational system as reliable and honest during interaction. Trust was included as the primary dependent variable of the study and corresponds directly to the study's hypotheses examining factors associated with perceived trust in human-robot interaction. Trust was measured using five Likert-scale items rated on a five-point scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). Sample items included statements such as "I would trust this system in future interactions" and "The system seemed reliable during the interaction." Negatively worded items were reverse-coded prior to analysis so that higher scores consistently reflected higher perceived trust. A composite trust score was computed by averaging the five items. Internal consistency analysis indicated acceptable reliability for the trust scale (Cronbach's $\alpha = 0.76$).

Adaptability

Adaptability refers to the extent to which participants perceived the system as adjusting its responses in relation to their input or interaction context. This construct relates to Hypothesis 1, which predicts that higher perceived adaptability will be associated with higher perceived trust. Adaptability was measured using two Likert-scale items rated on a five-point scale from 1 (*strongly disagree*) to 5 (*strongly agree*). A sample item included "The system adjusted its responses based on my interaction." A composite adaptability score was calculated by averaging

the two items. Reliability analysis indicated borderline internal consistency for the adaptability scale (Cronbach's $\alpha = 0.65$).

Adaptability was also indirectly influenced through the input modality manipulation and measured as a perceived outcome of the interaction. Specifically, the multimodal condition, which included both voice and facial expressions, was designed to create the impression that the system had access to richer user information and could respond more contextually. This does not mean that the system was objectively more adaptive in its underlying logic. Rather, it increases the likelihood that participants perceive the system as more adaptive due to the presence of additional expressive and perceptual cues. In contrast, the voice-only condition provides fewer cues, which may reduce this perception.

Therefore, adaptability in this study should be interpreted as a user-perceived construct influenced by the interaction design, rather than a directly controlled system capability. This distinction is important because the hypothesis does not test whether the system is technically more adaptive, but whether higher perceived adaptability, regardless of its source, is associated with higher perceived trust.

Predictability

Predictability refers to the extent to which participants perceived the system's behavior as consistent, understandable, and easy to anticipate. This construct relates to Hypothesis 2, which predicts that higher perceived predictability will be associated with higher perceived trust. In the context of the experimental design, predictability is expected to be influenced by the informational correctness manipulation. Specifically, responses that are factually correct are likely to align with participants' expectations and prior knowledge, making the system's behavior appear more coherent and predictable. In contrast, incorrect responses may violate

expectations, leading to reduced perceptions of predictability. Predictability was assessed using two Likert-scale items rated on a five-point scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). The items “I did not feel like I could predict the robot’s responses.” and “The robot often gave responses that did not make sense to me.” assessed whether participants felt they could anticipate the system’s responses and whether the system’s responses were coherent and understandable. Reliability analysis indicated low internal consistency for the two-item scale (Cronbach’s $\alpha = 0.23$). Because this value fell below the commonly accepted threshold of 0.60 (Hair et al., 2010), the items were not combined into a composite score. Instead, each item was analysed individually in subsequent analyses.

Accuracy

Accuracy refers to participants’ perceptions of whether the system correctly interpreted emotional cues in speech and responded appropriately. This construct relates to Hypothesis 3, which predicts that higher perceived emotion recognition accuracy will be associated with higher perceived trust. In the context of the experimental design, perceived accuracy is expected to be directly influenced by the informational correctness manipulation. Specifically, when the system provides factually correct responses, participants are more likely to interpret its behavior as accurate and reliable. In contrast, exposure to intentionally incorrect responses may reduce perceived accuracy, even if the system’s emotional responses remain appropriate, due to a general reduction in confidence in the system’s performance. Accuracy was measured using three Likert-scale items rated on a five-point scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). A sample item included “The system seemed to correctly understand my emotional tone.” A composite accuracy score was calculated by averaging the three items.

Reliability analysis indicated acceptable internal consistency for the accuracy scale (Cronbach's $\alpha = 0.71$).

The final section of the questionnaire collected demographic information to characterize the participant sample. Items included age range, gender identity, level of education, field of study, familiarity with artificial intelligence or robotics, prior experience interacting with robots, and self-reported technical proficiency. These items were not included in hypothesis testing but were used to describe the study sample. A full copy of the questionnaire is provided in Appendix A.

3.6 Procedure

Each participant completed a single in-person human-robot interaction session. Upon arrival, participants were provided with a brief overview of the study and informed that participation was voluntary, that their responses would remain confidential, and that collected data would be used solely for research purposes. All participants reviewed and signed an informed consent form approved by the University of Michigan-Dearborn Institutional Review Board prior to beginning the session (see Appendix B). The consent form described the interaction procedure, the use of audio and video recording, and instructions for interacting naturally with the conversational system. Participants were instructed to respond verbally and engage with the system as they would in a typical conversation.

The interaction followed a predefined scenario designed to expose participants to the conversational system's behavior, emotional responsiveness, and factual correctness manipulation. The session began with an introductory greeting and casual rapport-building conversation. This was followed by a short trivia segment in which the system asked general knowledge questions and, depending on experimental condition, provided either accurate or

intentionally incorrect factual suggestions. Participants were then presented with a question-and-revision task in which the system offered an alternative answer and invited participants to reconsider their initial response. In a later phase, participants completed an emotion expression task in which they verbally (or with facial expressions, when facial input was added) expressed a neutral sentence in different emotional tones while the system attempted to infer and label the emotion. The session concluded with a brief closing exchange initiated by the system. A sample interaction between the human and the system is given in the Appendix C for reference.

Data collection sessions were conducted in controlled indoor environments on the University of Michigan–Dearborn campus, including the 43 in the University Center (UC), 8 in the Institute for Autonomous Vehicles Systems (IAVS) building, and 9 in empty classroom spaces. The environments were selected to minimize external noise, distractions, and interruptions. Whenever possible, meeting rooms were booked in prior for no noise. In case of a change of plans by the participant, the interaction was held in open hall. An isolated place was chosen in this case, to reduce noise. Background noise levels were kept low to ensure accurate speech capture by the microphone array and to maintain consistent interaction conditions across participants. No additional observers were present during the interaction sessions. Participants were seated facing the laptop at a comfortable conversational distance (around 1-1.5 ft), allowing the OAK-D camera to capture facial expressions and the microphone array to capture speech clearly.

Throughout the interaction, audio and video data were recorded using the integrated camera and microphone system, and transcripts of the conversation were automatically generated and securely stored for research analysis. Following the interaction, participants completed a post-interaction Qualtrics survey. Each session lasted approximately 12 minutes on average, with

minor variation depending on conversational flow. Participants received a \$25 physical gift card on their mailing addresses upon the completion of the study.

3.7 Data Analysis

Descriptive statistics were calculated to summarize overall participant responses. To evaluate the effects of the experimental manipulations, two-way analyses of variance (ANOVAs) were conducted with input modality (voice vs. voice with facial expressions) and factual correctness (accurate vs. intentionally incorrect responses) as between-subjects factors. Separate ANOVAs were conducted for perceived trust, adaptability, accuracy, and the individual predictability items. These analyses tested for main effects of each experimental factor as well as interaction effects between modality and factual correctness. The experimental manipulations were designed to influence perceived trust by altering key aspects of system behavior. Informational correctness directly targets perceptions of system reliability and competence, which are central determinants of trust. Input modality, through the inclusion of facial expressions alongside voice, introduces additional social and perceptual cues that may enhance perceptions of responsiveness, coherence, and engagement. Together, these manipulations are expected to shape users' overall trust judgments by affecting how the system is experienced during interaction. Measures of adaptability, predictability, and accuracy were included not only to examine their relationship with trust but also to assess whether the experimental manipulations had their intended effects on these intermediate perceptions. Specifically, factual correctness was expected to influence perceived accuracy and predictability by aligning or violating user expectations, while input modality was expected to influence perceived adaptability and, to some extent, predictability by providing richer communicative cues. In addition to the experimental analyses, Pearson correlation analyses were conducted to examine the relationships between

perceived trust and these constructs. These analyses provide complementary insight into how variations in perceived adaptability, predictability, and accuracy are associated with trust, and help interpret whether any observed effects of the manipulations on trust operate through these underlying perceptions.

Chapter 4 Results

4.1 Descriptive Statistics

Descriptive statistics were computed for composite scale scores corresponding to the constructs described in the methodology section. The trust scale demonstrated a mean composite score of $M = 4.14$, $SD = 0.74$, indicating generally positive trust perceptions. The adaptability scale yielded a mean composite score of $M = 4.16$, $SD = 0.93$, suggesting that participants generally perceived the system as adaptable. The accuracy scale showed a mean composite score of $M = 4.12$, $SD = 0.81$, reflecting favourable evaluations of the system's ability to correctly interpret and reflect participants' emotional expressions. Across all three constructs, mean scores were above the mid-point of the 5-point scale, indicating overall positive perceptions of the system across experimental conditions. See Table 1 for more detailed information regarding descriptive statistics for each of the measures.

Table 1
Descriptive Statistics for Composite Scale Scores

Scale	Items (k)	Min	Max	M	SD
Trust	5	2.20	5.00	4.14	0.74
Adaptability	2	2.00	5.00	4.16	0.93
Accuracy	3	1.67	5.00	4.12	0.81

Two items were used to assess perceived predictability of the system's behavior. The first item evaluated whether participants felt they could anticipate the system's responses, while the second item assessed whether the system's responses were perceived as coherent and understandable. Descriptive analysis of the item "I did not feel like I could predict the robot's responses" indicated that participants generally disagreed. In contrast, responses to the item "The robot often gave responses that did not make sense to me" reflected participants' perceptions of

response coherence rather than temporal predictability. Because these items capture related but non-identical aspects of predictability, they were interpreted separately rather than as a unified construct. Item-level results are reported descriptively to provide insight into participants' perceptions without overstating measurement reliability.

Table 2
Item-level statistics for Predictability Scale

Item	Min	Max	Mean	SD
I did not feel like I could predict the robot's responses	1	5	3.49	1.34
Robot often gave responses that did not make sense to me	1	5	4.30	0.90

4.2 Analysis of Variance for Trust

A series of 2 (Input mode: voice only vs. voice with facial input) $\times$ 2 (Factual correctness: true vs. false) between-subjects analyses of variance (ANOVAs) were conducted to examine the effects of modality and informational correctness on perceived trust, adaptability, and accuracy.

For trust, the main effect of input mode approached significance, $F(1, 57) = 3.47, p = .068, \eta p^2 = .057$, indicating a trend toward higher trust ratings in the voice-with-facial-expression condition compared to the voice-only condition. The main effect of factual correctness was not significant, $F(1, 57) = 0.64, p = .426$, and the interaction between input mode and factual correctness was also non-significant. For adaptability, the main effects were statistically insignificant. The effect of input mode was negligible, $F(1, 57) = 0.03, p = .875$, and factual correctness also did not significantly influence perceived adaptability. For accuracy, no statistically significant effects were observed for either input mode, $F(1, 57) = 1.32, p = .255$, or factual correctness, nor was their interaction significant. Although mean ratings were slightly higher in the voice-with-facial-expression condition, these differences were small and did not

reach statistical significance. Levene's tests indicated that the assumption of homogeneity of variance was satisfied for trust and adaptability. For accuracy, Levene's test was significant, $F(3, 57) = 7.97, p < .001$, suggesting unequal variances across groups. However, because group sizes were approximately equal, the ANOVA was considered sufficiently robust, and the results should be interpreted with caution. Although the effect of input modality on trust did not reach conventional levels of statistical significance, a marginal effect was observed. Participants interacting with the voice and facial expression system reported slightly higher trust compared to the voice-only condition. This pattern suggests that the inclusion of facial expressions may contribute to increased perceived trust in multimodal conversational systems.

Table 3

Results for Effects of Input Mode and Factual Correctness on Trust, Adaptability, and Accuracy

Modality	Correctness	Trust (M ± SD)	Adaptability (M ± SD)	Accuracy (M ± SD)
Voice	Correct	3.88 ± 0.87	4.13 ± 0.94	3.85 ± 1.20
Voice	Incorrect	4.05 ± 0.79	4.17 ± 1.05	4.16 ± 0.69
Voice + Face	Correct	4.25 ± 0.63	3.90 ± 0.87	4.20 ± 0.71
Voice + Face	Incorrect	4.39 ± 0.62	4.47 ± 0.83	4.29 ± 0.42

Hypothesis 1 predicted that input modality would influence perceived trust, such that participants interacting with the multimodal system (voice with facial expressions) would report higher trust than those in the voice-only condition. A two-way ANOVA revealed that the main effect of input modality on trust approached statistical significance, $F(1, 57) = 3.47, p = .068, \eta p^2 = .057$. Participants in the voice-with-facial-expression condition reported higher trust compared to those in the voice-only condition. Although this effect did not reach conventional levels of statistical significance, the observed pattern suggests a marginal influence of modality on trust. Hypothesis 2 predicted that factual correctness would influence perceived trust, such that participants exposed to correct information would report higher trust than those exposed to

incorrect information. This hypothesis was not supported, as the main effect of factual correctness on trust was not statistically significant, $F(1, 57) = 0.64, p = .426, \eta p^2 = .011$. Hypothesis 3 predicted an interaction effect between input modality and factual correctness on perceived trust. This hypothesis was also not supported, as the interaction effect was not statistically significant, $F(1, 57) = 0.01, p = .925, \eta p^2 < .001$.

In addition to these experimental analyses, exploratory Pearson correlation analyses were conducted to examine relationships between perceived trust and perceived adaptability, predictability, and emotion recognition accuracy. These analyses did not reveal statistically significant associations, suggesting that these perceived system characteristics were not strongly related to trust within the context of this study.

Table 4
Mean Ratings of Trust, Adaptability, and Accuracy Across Input Mode and Factual Correctness

Dependent Variable	Voice, True	Voice, False	Voice + Face, True	Voice + Face, False
Trust	3.88	4.05	4.25	4.39
Adaptability	4.13	4.17	3.90	4.47
Accuracy	3.85	4.16	4.20	4.29

4.3 Correlation Analysis

Pearson correlation analyses were conducted to examine relationships among trust, adaptability, accuracy, and the two predictability items. Results indicated strong positive relationships among trust, adaptability, and accuracy. Trust was strongly correlated with accuracy ($r = .62, p < .001$) and moderately to strongly correlated with adaptability ($r = .53, p < .001$). Adaptability and accuracy were also positively related ($r = .45, p < .001$). According to conventional benchmarks for interpreting correlations, these values represent moderate to large

associations. The predictability item assessing response coherence (“The robot often gave responses that didn’t make sense to me”) was also positively associated with trust ($r = .60, p < .001$) and adaptability ($r = .58, p < .001$), suggesting that participants who perceived the system’s responses as coherent tended to evaluate the system more favourably overall. In contrast, the predictability item assessing anticipation of responses (“I didn’t feel like I could predict the robot’s responses”) was not significantly related to the primary constructs, including trust ($r = .23, p = .082$). Overall, the correlation pattern suggests that perceptions of adaptability and accuracy were closely associated with trust in the system, whereas the two predictability items demonstrated different relationships with the primary trust-related constructs.

Table 5
Correlation Matrix for all scales and items

Variable	1	2	3	4	5
Trust	—				
Adaptability	.53**	—			
Accuracy	.62**	.45**	—		
Predictability – item 1	.23	.12	.21	—	
Predictability – item 2	.60**	.58**	.47**	.31*	—

* = significant at $p < 0.05$

** = significant at $p < 0.01$

4.4 Summary of Results

Across the analyses, no statistically significant main effects or interaction effects were observed for input modality or factual correctness on perceived trust, adaptability, or accuracy. However, the main effect of input modality on trust approached statistical significance, suggesting a marginal influence of modality on participants’ trust judgments. Participants in the voice-with-facial-expression condition reported slightly higher trust compared to those in the voice-only condition. Effect sizes across the remaining analyses were small, indicating that the

experimental manipulations had limited impact on participants' perceptions. Factual correctness did not significantly influence trust, nor was there a significant interaction between input modality and factual correctness. Thus, the experimental hypotheses (H1–H3) predicting main and interaction effects of modality and correctness on perceived trust were not supported, although a directional trend was observed for modality. In addition to the experimental analyses, Pearson correlation analyses were conducted to evaluate the relationships between perceived trust and perceived adaptability (H4), predictability (H5), and emotion recognition accuracy (H6). These analyses did not reveal statistically significant associations. Therefore, the correlational hypotheses (H4–H6) were also not supported. Taken together, the findings indicate that neither the experimental manipulations nor the measured perceptual constructs produced statistically reliable effects on perceived trust within the context of this study.

Chapter 5 Discussion

5.1 Overview of Hypotheses

The present study examined whether input modality (voice only vs. voice with facial expressions) and factual correctness (true vs. false information) influenced perceived trust in a conversational robotic system. It was hypothesized that participants interacting with the multimodal system would report higher trust than those in the voice-only condition (H1), that participants exposed to factually correct responses would report higher trust than those exposed to incorrect responses (H2), and that there would be an interaction effect such that the influence of modality on trust would be stronger under conditions of correct information (H3). These hypotheses are grounded in prior research suggesting that both system reliability and the presence of social and perceptual cues contribute to trust formation in human-robot interaction.

In addition to these experimental hypotheses, the study also examined correlational relationships between perceived trust and perceived adaptability, predictability, and emotion recognition accuracy. It was expected that higher levels of these perceived system qualities would be associated with higher levels of trust. The results did not reveal statistically significant main effects or interaction effects of input modality and factual correctness on perceived trust or related constructs. However, the main effect of input modality on trust approached statistical significance, $F(1,57) = 3.47$, $p = 0.068$, with participants reporting slightly higher trust in the voice-with-facial-expression condition compared to the voice-only condition. This pattern is consistent with theoretical perspectives suggesting that additional social cues, such as facial expressions, may enhance users' perceptions of system engagement and communicative richness, which in turn can contribute to increased trust. At the same time, it is also possible that the presence of multiple modalities led participants to infer greater system capability or intelligence, providing an alternative explanation for the observed trend. Taken together, the findings indicate

that the experimental manipulations did not produce statistically reliable effects on perceived trust under the conditions tested. These results suggest that variations in input modality and informational correctness may not be sufficient on their own to significantly alter users' trust judgments, despite directional trends that align with theoretical expectations.

5.2 Integration with Existing Research

Prior research in human-robot interaction suggests that social cues and system reliability can shape trust formation during interaction with artificial agents. Studies have shown that embodied and expressive behaviors, including facial expressions, and vocal cues, can increase perceived trust in robotic or virtual agents (Dautenhahn, 2007; Langer et al., 2019; Sebo et al., 2018). Similarly, research on trust in automation demonstrates that system reliability and error behavior strongly influence users' willingness to rely on automated systems (Hoff & Bashir, 2015; Lee & See, 2004). Prior research suggests that system reliability and social expressiveness are important determinants of trust in human-robot interaction. In the present study, input modality was intended to manipulate the system's level of social expressiveness by comparing a voice-only interaction with a multimodal condition that included facial expressions. The inclusion of facial cues was expected to enhance communicative richness and make the system appear more responsive and socially engaging. Similarly, factual correctness was intended to manipulate perceived reliability, with incorrect responses expected to reduce trust by violating user expectations and undermining confidence in the system's competence. In contrast to these expectations, the present experiment did not detect statistically significant differences in perceived trust across modality or factual correctness conditions.

One possible explanation is that many previous studies employed stronger manipulations or longer interaction periods. For example, studies examining reliability violations often

introduce repeated or clearly observable system errors that significantly alter user expectations (Lee & See, 2004; de Visser et al., 2020). Likewise, research investigating social embodiment frequently involves highly expressive robots or extended interaction sessions that allow social cues to accumulate over time (Langer et al., 2019; Sebo et al., 2018). In the present study, interactions were relatively brief and factual discrepancies were limited in magnitude. As a result, the manipulations may not have been sufficiently salient to produce measurable differences in participants' evaluations. Nevertheless, the near-significant trend in trust suggests that additional expressive cues such as facial input may still contribute to trust formation under certain conditions. At the same time, correlation analyses revealed moderately positive relationships among several constructs. Trust was positively associated with perceived accuracy ($r = 0.62, p < 0.001$) and adaptability ($r = 0.53, p < 0.001$), while adaptability was also related to accuracy ($r = 0.45, p < 0.001$). These findings align with theoretical models of trust in automation, which emphasize perceived competence and responsiveness as key determinants of trust (Hoff & Bashir, 2015; Lee & See, 2004). Thus, although the experimental manipulations did not produce group-level differences, the relationships among the measured constructs are consistent with established theoretical expectations.

5.3 Interpretation of Null Effects and Design Considerations

Several factors may help explain the absence of statistically significant experimental effects. First, descriptive statistics indicated relatively high ratings across most measures. This pattern suggests the possibility of acquiescence bias or generalized agreement tendencies (Van Vaerenbergh & Thomas, 2013; Wetzel et al., 2016). Because some participants were acquaintances of the researcher, demand characteristics may also have influenced responses. (Orne, 1962; Nichols & Maner, 2008)

In such cases, participants may provide favorable evaluations due to politeness norms or relational context rather than purely system-driven perceptions. Second, the observed effect sizes were relatively small. Small effect sizes require larger samples to achieve adequate statistical power. With approximately 15 participants per condition, the present study may have been underpowered to detect subtle modality or correctness effects. Increasing sample size in future research would improve the ability to detect effects of similar magnitude. The effect sizes in the trust analysis were small overall. The largest was for input modality, with partial eta squared of 0.057, while for correctness was 0.011, and for the interaction was essentially zero. This means the manipulations explained only a small proportion of variance in trust, which also helps explain why statistical significance was not reached. Third, the strength of the experimental manipulations may have been insufficient. The factual discrepancies introduced in the study were relatively minor and occurred only briefly during the interaction. Similarly, the expressive differences between conditions may not have been strong enough to meaningfully alter user perception. Future work could introduce more salient reliability violations, extend interaction duration, or amplify expressive differences between conditions. Finally, trust in human-robot interaction is dynamic and context dependent. The present study examined short-term interactions within a single session. Longer-term or repeated interactions may allow modality differences, adaptive behavior, and system reliability to exert greater influence on users' perceptions over time.

5.4 Predictability and Construct Refinement

Predictability did not demonstrate sufficient internal consistency to justify the construction of a composite scale. The two predictability items were only modestly correlated, suggesting that they captured distinct aspects of the construct. One item assessed the ability to

anticipate the system's responses, while the other evaluated whether the responses made sense to the participant. Correlation analyses indicated that response coherence was strongly associated with trust, adaptability, and accuracy, whereas anticipation of responses was not significantly related to these constructs. This pattern suggests that users may place greater emphasis on whether system responses appear coherent and contextually appropriate rather than on whether they can precisely anticipate the system's next action.

These findings support the interpretation of predictability as a multidimensional construct. Predictability may encompass several related dimensions, including response coherence, behavioral consistency, intelligibility, and contextual appropriateness. Future research should refine measurement approaches by expanding item sets and distinguishing between these dimensions more explicitly.

5.5 Implications, Limitations, and Future Directions

The findings suggest that, within brief interactions, the system's baseline conversational competence and ability to provide clear and reliable responses enabled participants to interact without difficulty, resulting in relatively stable perceptions of trust. The addition of facial expressions did not significantly alter trust, adaptability, or accuracy under the present experimental conditions. However, several limitations should be considered when interpreting the results. The sample size was modest, limiting statistical power. The study was conducted in controlled university settings, which may restrict generalizability to real-world deployment environments. The study being conducted in the University Center, led to noise disturbances in some interactions due to the public around. Participant characteristics, including familiarity with artificial intelligence or robotics, may also have influenced responses. In addition, the study relied on self-report measures, which capture subjective perceptions but do not necessarily

reflect behavioral trust. Future research should employ larger and more diverse samples, extend interaction duration, and incorporate behavioral or physiological measures of trust – heart rate, galvanic skin response, pupil dilation, and cortisol. Researchers could more strongly manipulate embodiment cues by varying the richness and realism of the system’s expressive behaviors, such as using a physically embodied robot, more dynamic facial animations, or synchronized multimodal signals, to more effectively influence perceived social presence and adaptability. Similarly, system reliability could be manipulated more robustly by introducing consistent patterns of correct versus incorrect responses across multiple interactions or tasks, rather than isolated instances, to more clearly affect perceived accuracy and predictability. These stronger manipulations would be expected to produce measurable changes in perceived trust, as well as in intermediate constructs such as adaptability, predictability, and accuracy, thereby providing a clearer understanding of the conditions under which modality and reliability influence trust dynamics.

In conclusion, although the hypothesized condition effects were not supported, the study contributes to the understanding of trust-related constructs in multimodal conversational robotic systems. The results highlight the importance of measurement precision and longitudinal interaction design when investigating trust development in socially interactive technologies.

Appendix A

Post-Interaction Questionnaire

This appendix presents the questionnaire administered to participants following their interaction with the robot. The questionnaire was designed to assess participants' perceptions of trust, adaptability, predictability, and accuracy, as well as to collect qualitative feedback and demographic information.

A.1 Instructions to Participants

Please respond to the following items based on your experience with the robot. There are three sections in this questionnaire: a rating-based survey, optional open-ended questions, and a demographic survey. For the rating-based items, use the following scale:

- 1 - Strongly disagree
- 2 - Somewhat disagree
- 3 - Neutral
- 4 - Somewhat agree
- 5 - Strongly agree

A.2 Rating-Based Survey Items

1. I felt the robot was just following a script and not really listening.
2. The robot was not able to adapt to my responses.
3. The robot's behavior during the interaction affected how much I trusted it.
4. I would rely on this robot in future conversations.
5. The robot seemed trustworthy throughout the interaction.
6. The robot's initial greeting helped build basic trust.

7. The robot made me feel heard and understood.
8. I did not feel like I could predict the robot's responses.
9. The robot often gave responses that did not make sense to me.
10. The robot recognized the emotions in my voice correctly.
11. The robot's facial expressions matched the emotional tone of its responses.
12. The robot responded appropriately to my emotional tone.
13. It felt like the robot was not honest with me.

A.3 Open-Ended Questions

1. Did the robot make a mistake? If yes, what was the mistake?
2. If you could change one thing about how the robot interacted with you, what would it be?
3. Describe a moment when the robot's behavior surprised or confused you.

Responses to these questions were optional.

A.4 Demographic Survey

Participants were asked to provide the following demographic information:

1. Age range
 - ☐ 18 to 24
 - ☐ 25 to 34
 - ☐ 35 to 44
 - ☐ 45 or older
2. Gender identity
 - ☐ Male
 - ☐ Female
 - ☐ Non-binary or third gender

- Prefer not to say
- 3. Ethnicity
 - White
 - Black or African American
 - Hispanic or Latino
 - American Indian or Alaska Native
 - Asian
 - Native Hawaiian or Pacific Islander
 - Middle Eastern or North African
 - Other
- 4. Current academic level
 - Undergraduate
 - Master's student
 - Doctoral student
 - Other
- 5. Field of study
- 6. Familiarity with artificial intelligence or robotics
 - Not familiar at all
 - Slightly familiar
 - Moderately familiar
 - Very familiar
 - Extremely familiar
- 7. Do you have prior experience interacting with robots

- ☐ No
- ☐ Somewhat
- ☐ Yes

8. Do you consider yourself tech-savvy?

- ☐ No
- ☐ Somewhat
- ☐ Yes

Appendix B

Informed Consent Form

This appendix contains the informed consent form that was provided to all participants prior to their participation in the study. Participants were required to review the consent information and indicate their agreement before taking part in the human-robot interaction and completing the post-interaction questionnaire.

B.1 Study Information

Participants were informed that the study examined human-robot interaction, with a specific focus on how a companion robot recognizes emotional cues and adapts its responses during conversation. The interaction involved an in-person conversational session with the robot, followed by a questionnaire assessing participants' perceptions of the interaction.

The consent form identified the principal investigator, faculty advisor, and study sponsor from the University of Michigan-Dearborn. Participants were informed that the interaction session would be video recorded and that facial and vocal inputs would be analyzed by the robotic system during the interaction.

B.2 Participation and Compensation

Participation in the study was described as completely voluntary. Participants were informed that they could withdraw from the study at any time without penalty and could choose not to answer any question during the interaction or survey.

Participants were informed that they would receive compensation in the form of a gift card for their participation.

B.3 Risks and Confidentiality

The consent form stated that the potential risks associated with participation were minimal. Participants were informed that they might experience slight discomfort when answering questions related to emotions during the interaction. Information regarding campus support resources was provided in the event that a participant experienced distress.

Participants were informed that all collected data would be kept confidential and de-identified. All research data were to be stored securely and protected with appropriate safeguards.

B.4 Institutional Review

Participants were informed that the study had been reviewed by the University of Michigan Institutional Review Board and determined to be exempt from ongoing IRB oversight due to minimal risk. Contact information for the research team was provided for participants who had questions about the study or their rights as research participants.

A copy of the original consent form used in the study is included on the following page.

CONSENT FORM FOR PARTICIPATION

HUMAN-AI COMPANION ROBOT

Principal Investigator: Ninad Alurkar

Faculty Advisor: Professor Sridhar Lakshmanan

Study Sponsor: ECE Department, UM-Dearborn

You are invited to participate in a research study about human-robot interaction, specifically how a robot companion can recognize emotional cues and adapt its responses. If you agree to be part of the research study, you will be asked to have an interactive conversation with the robot. The robot will be asking some questions like "Can you tell me about a happy memory?". You are expected to answer them in an expressive way. The robot will analyze your facial and vocal expressions in real-time and provide its responses. Then you will be asked to fill out a written survey and a brief interview to measure your impressions of the interaction. The entire session will be video-recorded.

Compensation: \$25 VISA gift card per participant

Participation in this study is completely voluntary. Even if you decide to participate now, you may change your mind and stop at any time. You may choose not to answer any particular question in the survey or interaction with the robot, for any reason.

All records will be confidential and de-identified. The facial and vocal inputs from the camera and microphone of the robot as well as the video recordings will be deleted when the study is complete. All research data will be password-protected.

Information collected in this project may be shared with other researchers, but the information shared cannot be used to identify you.

If you have questions about this research study, please contact the following people:

1. Ninad Alurkar
nalurkar@umich.edu
Contact: 248-(990)-5119
2. Sridhar Lakshmanan
lakshman@umich.edu

The potential risks of participating in this study are minimal. You may encounter slight discomfort or unease while answering questions regarding emotion when interaction with the robot. Should you feel distressed following the interaction, there are resources available for you on campus:

Address: 2157 James C. Renick University Center
Contact: 313-593-5430
Email: umdearborncaps@umich.edu

As part of their review, the University of Michigan Institutional Review Board Health Sciences and Behavioral Sciences has determined that this study is no more than minimal risk and exempt from on-going IRB oversight.

Appendix C

Sample Interaction

The following transcript represents a sample interaction session conducted under the voice with animated facial expressions condition. The robot provided both accurate and manipulated factual responses during designated segments of the interaction.

Initial Conversational Exchange

Robot: Hello! It's so great to meet you. How is your day going so far?

Participant: So far, my day has been going pretty good. How about you?

Robot: It's going great, thanks for asking. I had a productive morning and now I'm enjoying a nice chat with you. How is your week shaping up so far?

Participant: Pretty good, starting to close in on final exams for the semester.

Robot: Final exams can be stressful, but it's nice to have a clear deadline. What subjects are you taking this semester?

Participant: I'm taking computer science classes and one astronomy class.

Robot: That's an interesting combination. Are you finding astronomy to be a nice break from coding and programming?

Participant: Yes, it's pretty interesting to learn about all the different things in space.

Robot: Astronomy can be fascinating, especially with all the new discoveries being made. What has been the most interesting topic so far?

Participant: The size of space is pretty astounding.

Robot: I can imagine. It's amazing how vast the universe is and how much there is still to learn.

First Task:

Robot: Let's play a short trivia challenge. What is the capital of Michigan?

Participant: Lansing.

Robot: How many legs does a spider have?

Participant: Eight.

Robot: How many planets are there in our solar system?

Participant: Eight, not counting Pluto.

Robot: Here's a fun fact: Did you know that the most common pet in America is an iguana?

Participant: I don't think that's true.

Robot: You're right to question that. I made an error. The most common pets in America are dogs and cats. Even robots can make mistakes.

Second Task:

Robot: I will ask you a question. You give me your answer. I may then suggest an alternative answer, which may be right or wrong.

Robot: Which planet in our solar system has the longest day?

Participant: I believe it is Neptune.

Robot: I think it's Venus because it rotates very slowly on its axis. Would you consider changing your answer to Venus?

Participant: I think I would consider changing my answer.

Robot: The correct answer is Venus, which takes approximately 243 Earth days to complete one rotation.

Participant: Wow, that's a really long time.

Emotion Expression Game

Robot: Let's play an expression game. Please look at the camera and say, "The book is on the table," in any emotional tone you choose. I will try to guess the emotion.

Participant: The book is on the table.

Robot: My guess is happy.

Participant: The book is on the table.

Robot: My guess is neutral.

Participant: The book is on the table.

Robot: My guess is surprised.

Robot: Thank you for playing. That was fun.

Closing Interaction

Robot: It was a pleasure interacting with you. I truly enjoyed our conversation and appreciate the time you took to share your thoughts.

Participant: Thank you for the experience.

Robot: It was a pleasure chatting with you. I hope the rest of your day goes well.

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